Accuracy and Imbalance

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

One readable imbalance-comparison record.

Trace counts and class recalls, compare summaries and name the unresolved consequence decision.

Keep one habit: W1–W3/W5 → selected100-case task → project metric record → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Try the **recall** before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Confusion Matrix. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Train a Classifier. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Baselines First. Give an example and a mistake that the idea helps you avoid.

Confusion Matrix

> **My remembered idea, example and correction**

Train a Classifier

> **My remembered idea, example and correction**

Baselines First

> **My remembered idea, example and correction**

W1 · Explain the mechanism

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main training/evaluation has16clear/4blocked,blockedpositive. Baseline clear matrix[[16,0],[4,0]]:ordinary80%,class recalls100%/0%,balanced50%. Candidate[[12,4],[0,4]]:ordinary80%,recalls75%/100%,balanced87.5%,FP4/FN0. Changed8/8fixture baseline[[8,0],[8,0]]ordinary50%,candidate[[6,2],[0,8]]87.5%;balanced summaries unchanged by construction.

  Separate pictured18healthy/2spotted candidateTP2,FN0,FP3,TN15:85%accuracy/100%spotted recall,baseline90%/0%. Metrics describe different tradeoffs, not a safety guarantee.

- Reference for this exercise: open this week's student lesson, "Accuracy and Imbalance", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- **Training** has 16 clear and 4 blocked labels. State the majority rule. On the worked evaluation, give its matrix, **accuracy**, false positives and false negatives with blocked positive. Explain why 80 percent is incomplete.

> **My reasoning and evidence**

W2 · Trace the current experiment

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main training/evaluation has16clear/4blocked,blockedpositive. Baseline clear matrix[[16,0],[4,0]]:ordinary80%,class recalls100%/0%,balanced50%. Candidate[[12,4],[0,4]]:ordinary80%,recalls75%/100%,balanced87.5%,FP4/FN0. Changed8/8fixture baseline[[8,0],[8,0]]ordinary50%,candidate[[6,2],[0,8]]87.5%;balanced summaries unchanged by construction.

  Separate pictured18healthy/2spotted candidateTP2,FN0,FP3,TN15:85%accuracy/100%spotted recall,baseline90%/0%. Metrics describe different tradeoffs, not a safety guarantee.

- Reference for this exercise: open this week's student lesson, "Accuracy and Imbalance", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- For the worked candidate [[12,4],[0,4]], give ordinary **accuracy**, both **class-specific correct fraction**s, **balanced accuracy**, false positives and false negatives. Compare with the baseline.

> **My reasoning and evidence**

W3 · Repair the reasoning

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main training/evaluation has16clear/4blocked,blockedpositive. Baseline clear matrix[[16,0],[4,0]]:ordinary80%,class recalls100%/0%,balanced50%. Candidate[[12,4],[0,4]]:ordinary80%,recalls75%/100%,balanced87.5%,FP4/FN0. Changed8/8fixture baseline[[8,0],[8,0]]ordinary50%,candidate[[6,2],[0,8]]87.5%;balanced summaries unchanged by construction.

  Separate pictured18healthy/2spotted candidateTP2,FN0,FP3,TN15:85%accuracy/100%spotted recall,baseline90%/0%. Metrics describe different tradeoffs, not a safety guarantee.

- For the illustrative 8-clear 8-blocked set, use baseline [[8,0],[8,0]] and candidate [[6,2],[0,8]]. Calculate ordinary and **balanced accuracy** for each. Explain what stayed fixed and what this example does not prove.

> **My reasoning and evidence**

MAIN LESSON · SECTION 16

W5 · Plan the next check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main training/evaluation has16clear/4blocked,blockedpositive. Baseline clear matrix[[16,0],[4,0]]:ordinary80%,class recalls100%/0%,balanced50%. Candidate[[12,4],[0,4]]:ordinary80%,recalls75%/100%,balanced87.5%,FP4/FN0. Changed8/8fixture baseline[[8,0],[8,0]]ordinary50%,candidate[[6,2],[0,8]]87.5%;balanced summaries unchanged by construction.

  Separate pictured18healthy/2spotted candidateTP2,FN0,FP3,TN15:85%accuracy/100%spotted recall,baseline90%/0%. Metrics describe different tradeoffs, not a safety guarantee.

- An evaluation contains ten clear cases and no blocked cases. All **prediction**s are clear. State ordinary **accuracy**, explain why blocked performance is unavailable, and describe how our code handles the missing class.

> **My reasoning and evidence**
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MAIN LESSON · SECTION 4

Try alone · Selected experiment task

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- This selected task has90healthy/10spotted and its own candidate TP7,FN3,FP8,TN82. Use spotted as the declared positive class and retain these100cases as the denominator. No actual consequence weights or broad reliability evidence are supplied; give conditional reasoning rather than declaring a universally best model. Do not substitute the20-case main or pictured matrices.

- Validation has90 healthy and10 spotted cases. Always-healthy predicts all healthy. Another model has TP7,FN3,FP8,TN82. Compare **accuracy** and spotted **recall**.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Changed independent case

> **My independent explanation, trace and limits**
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MAIN LESSON · SECTION 9

Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Check Plant Classifier class balance and select metrics that keep the spotted-leaf class visible.

- The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use W2: Compare both class fractions for **Accuracy** and **class imbalance**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use the worked case for **Accuracy** and **class imbalance**. Proposed explanation: "High **accuracy** means both classes work well". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

Reference · Earlier learning target

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Model A is always healthy on 96 healthy and 4 spotted leaves. Model B finds 3 spotted but falsely flags 5 healthy. Compare **accuracy** and spotted detection.

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A dataset has 95 healthy and 5 spotted. Model predicts healthy for all.
- Calculate and critique **accuracy**.
- Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · Changed case

REFERENCE · SECTION 7

Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A classifier predicts healthy for all 100 plants; 95 are healthy and 5 diseased. Compute **accuracy** and diseased **recall**.
- My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 16 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Model A is always healthy on 96 healthy and 4 spotted leaves. Model B finds 3 spotted but falsely flags 5 healthy. Compare **accuracy** and spotted detection.

- Record support used: none, hint or teacher explanation.

My independently reasoned answer

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- New **training** labels have a clear majority. A separate evaluation has 18 clear and 2 blocked records.

  The always-clear baseline is [[18,0],[2,0]] and candidate is [[15,3],[1,1]], with reference rows and clear then blocked on both axes. Calculate ordinary **accuracy**, both class fractions and **balanced accuracy** for each.

  Give candidate FP and FN, then explain why the metrics do not choose an automatic winner.

My reasoning and evidence

▼ Extra practice · Pictured case

REFERENCE · SECTION 18

Extra practice · Pictured case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Show how a common healthy class can hide complete failure to detect spotted leaves.
- Toy evaluation has 18 healthy and 2 spotted leaves. Always-healthy predicts all healthy. A candidate hasTP2, FN 0, FP 3, TN 15.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- Why can the 90% baseline still be unsuitable for a detection task? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-4/16

Candidate decisions

	predicted spotted	predicted healthy
actual spotted	2	0
actual healthy	3	15

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/16

Accuracy and Imbalance

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Score trivial baseline

Always healthy is 18/20=90% accurate but detects0/2 spotted leaves.



2. Score candidate

Candidate accuracy is 17/20=85%; spotted recall is 2/2=100%.



3. Compare consequences

Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs.

CHANGE ONE THING

The candidate's three false alarms are corrected without changing its detections.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 19

Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Project: Responsible Plant Classifier
- Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 20

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Show how a common healthy class can hide complete failure to detect spotted leaves.
- Which labelled step matches this detail? Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs.
- 1. Compare consequences
- 2. Score candidate
- 3. Score trivial baseline
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Compare consequences Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs. Read the picture in order.

Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: The candidate’s three false alarms are corrected without changing its detections. Which conclusion is supported by the supplied facts?
- 1. Candidate **accuracy** becomes20/20 with spotted **recall** still100% in this toy set.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Candidate **accuracy** becomes20/20 with spotted **recall** still100% in this toy set. False alarms move from FP to TN; neither TP nor FN changes.

Repair and keep the record

Review Confusion Matrix (ai-4/15). Goal: Build a labelled confusion matrix from matched reference and **prediction** pairs. Short practice: Why is adding TP alone without reducing FN an incorrect revision?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Train a Classifier (ai-4/13). Goal: Connect **training**-only scaling and neighbour **prediction** in one pipeline. Short practice: Which pipeline record would help another team reproduce the result?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Baselines First (ai-4/10). Goal: Fit a majority label baseline using **training** labels only. Short practice: Can the candidate's **training accuracy** replace this validation comparison?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

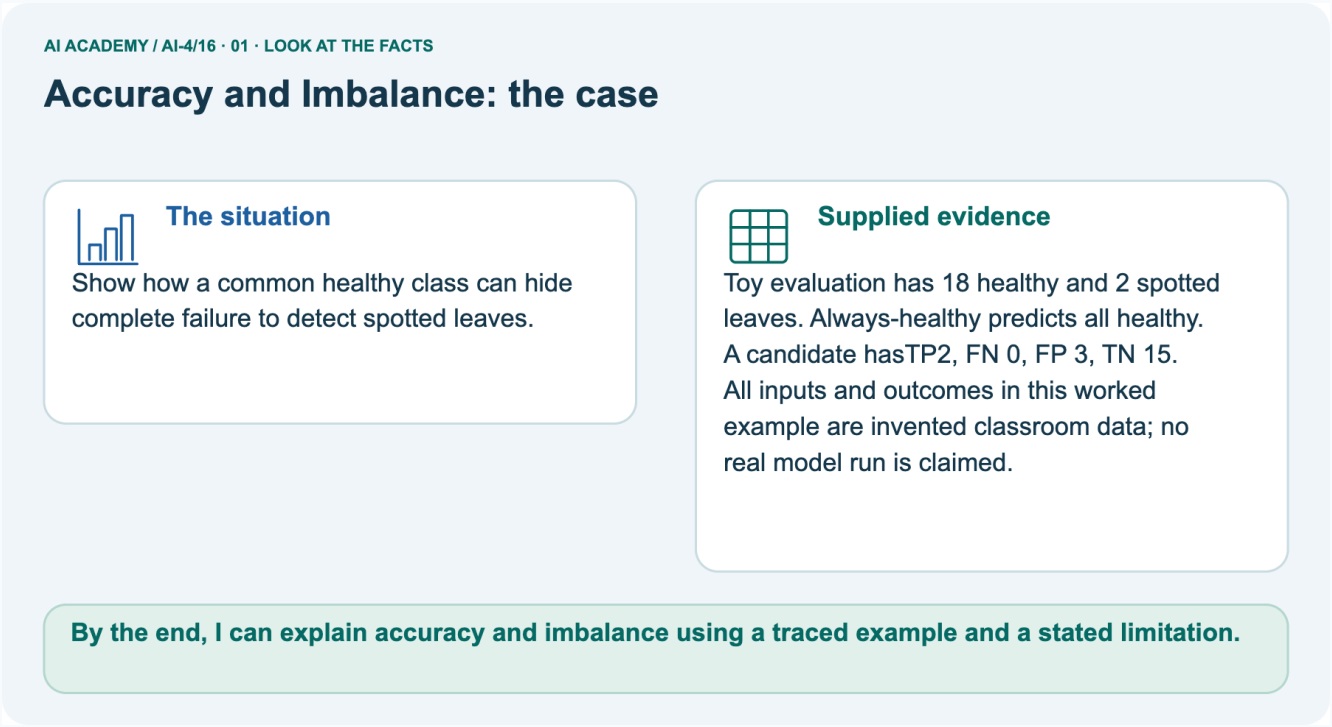
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the relevant picture instead of repeating every task.

01. Accuracy and Imbalance: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

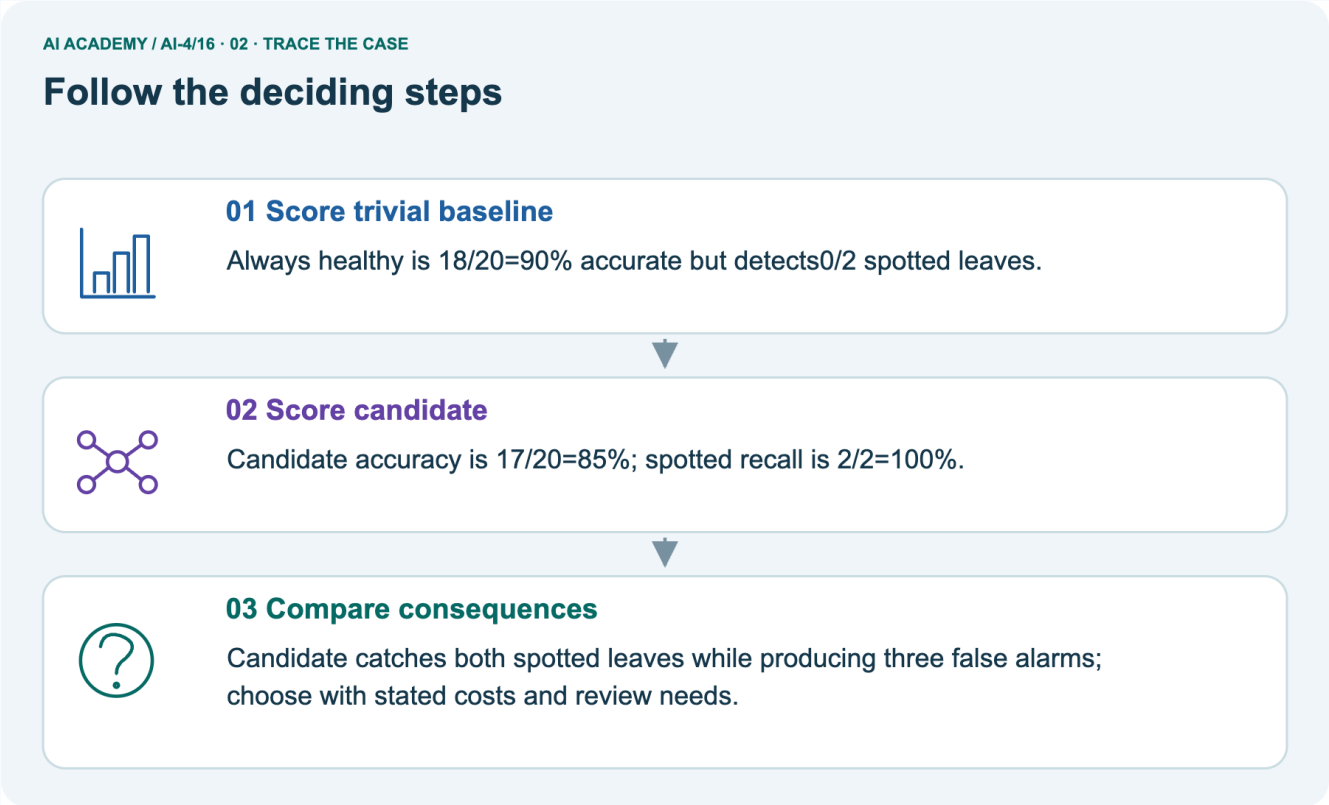
Show how a common healthy class can hide complete failure to detect spotted leaves.
Supplied facts: Toy evaluation has 18 healthy and 2 spotted leaves. Always-healthy predicts all healthy.

A candidate hasTP2, FN 0, FP 3, TN 15. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



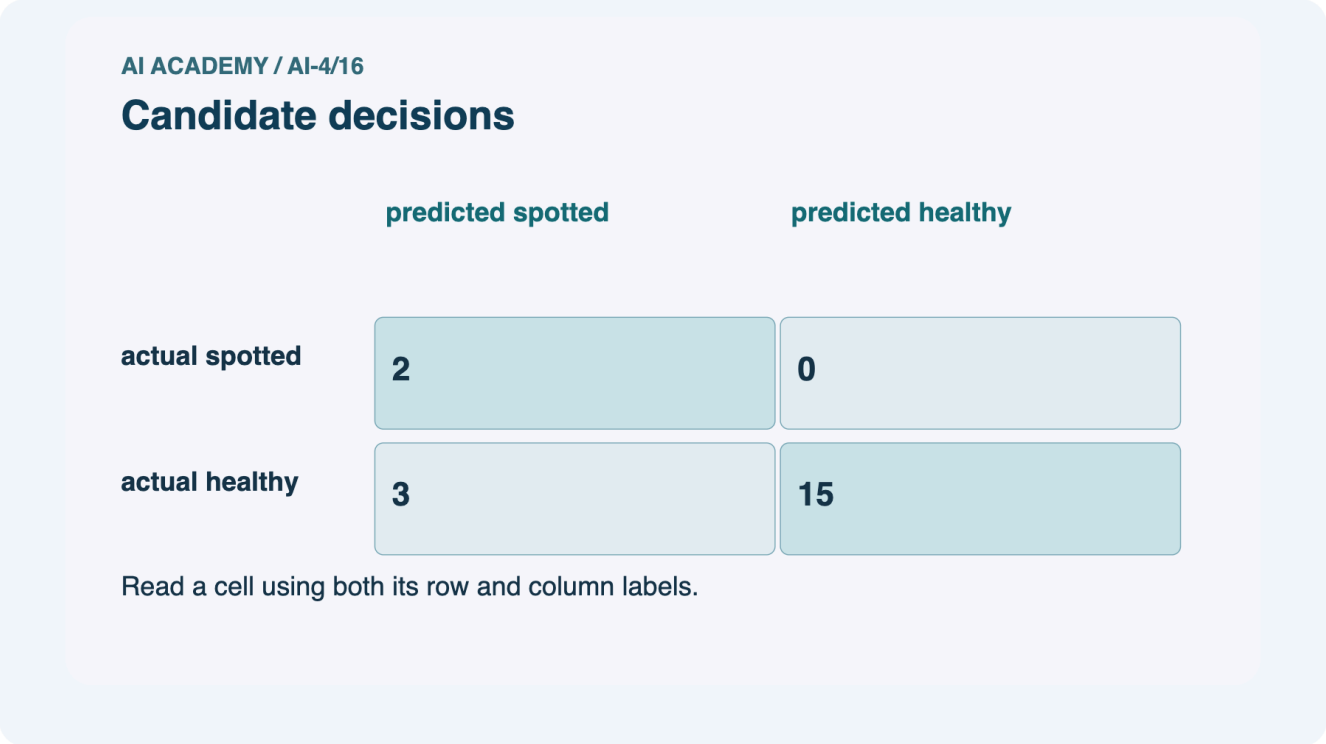
Follow the labelled picture with the matching explanation. Open the vector diagram

Score trivial baseline: Always healthy is 18/20=90% accurate but detects0/2 spotted leaves. → Score candidate: Candidate **accuracy** is 17/20=85%; spotted **recall** is 2/2=100%. → Compare consequences: Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Candidate decisions



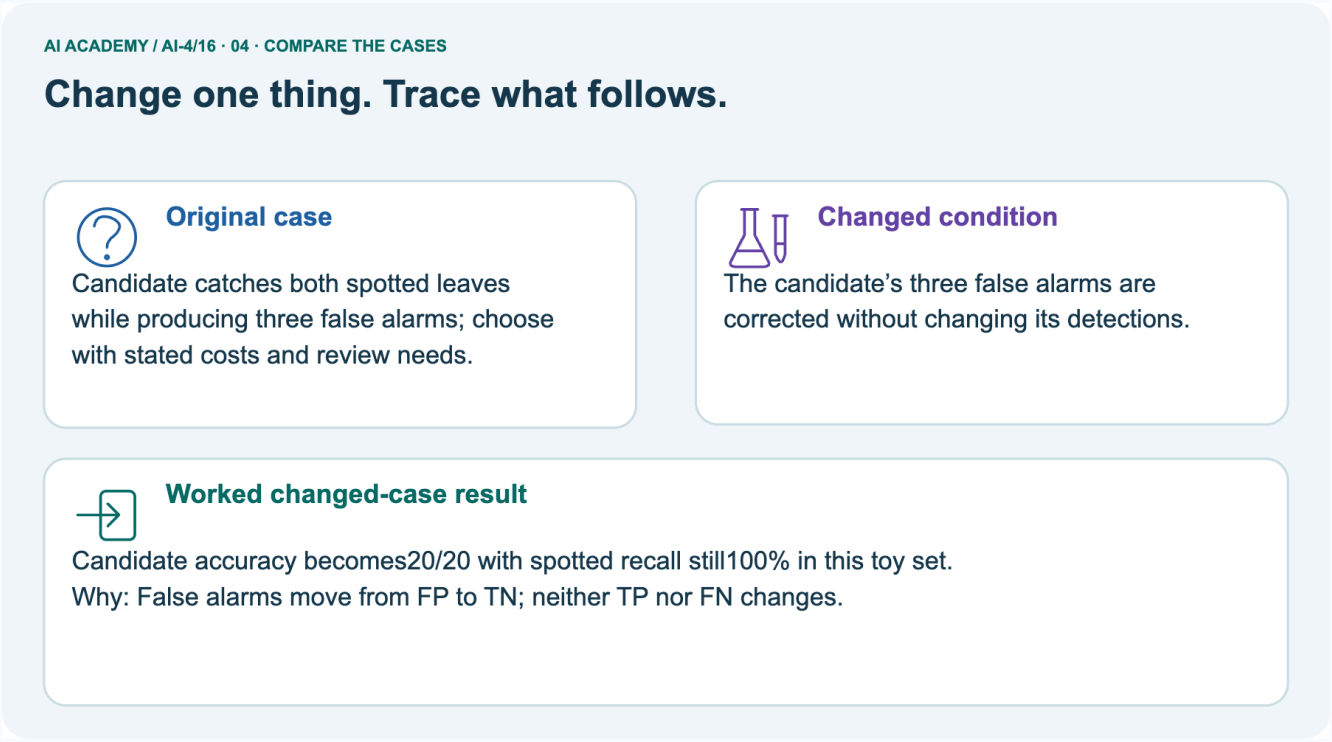
Follow the labelled picture with the matching explanation. Open the vector diagram

Candidate decisions. actual spotted: predicted spotted = 2, predicted healthy = 0. actual healthy: predicted spotted = 3, predicted healthy = 15.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs.

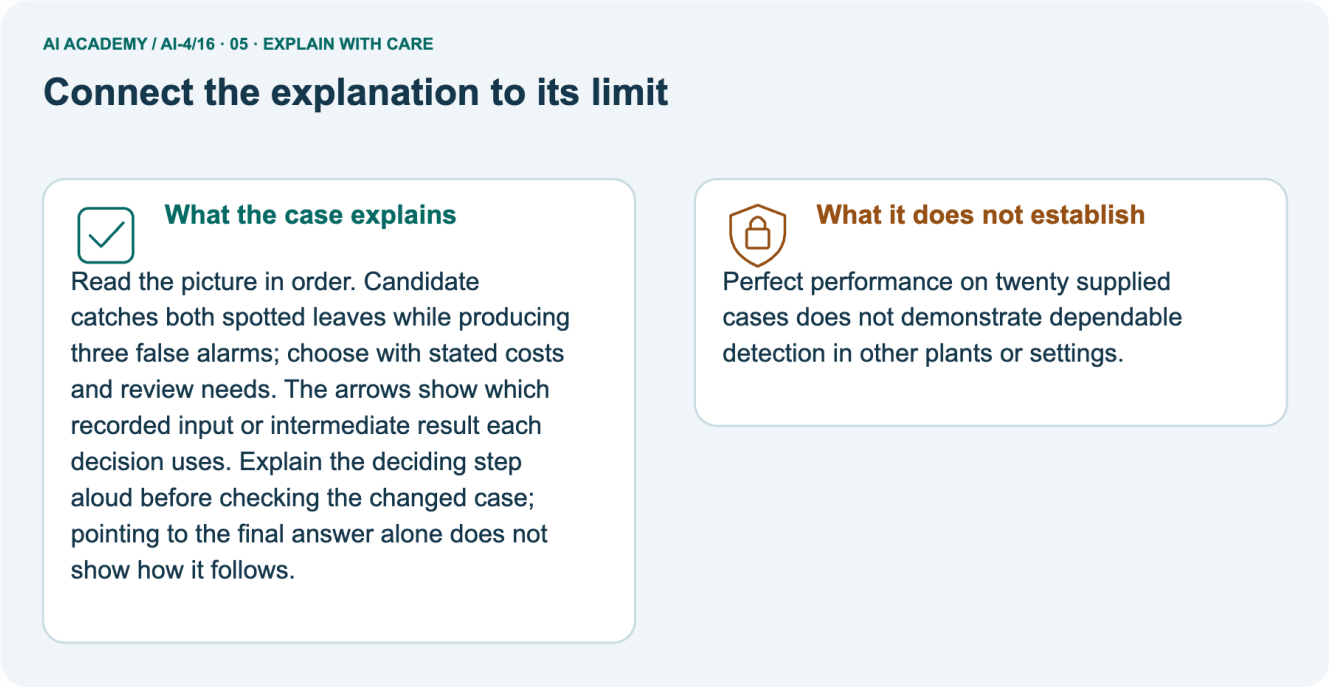
Changed condition: The candidate's three false alarms are corrected without changing its detections. Expected result: Candidate **accuracy** becomes20/20 with spotted **recall** still100% in this toy set.

Reason: False alarms move from FP to TN; neither TP nor FN changes.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs. The arrows show which recorded **input** or intermediate result each decision uses.

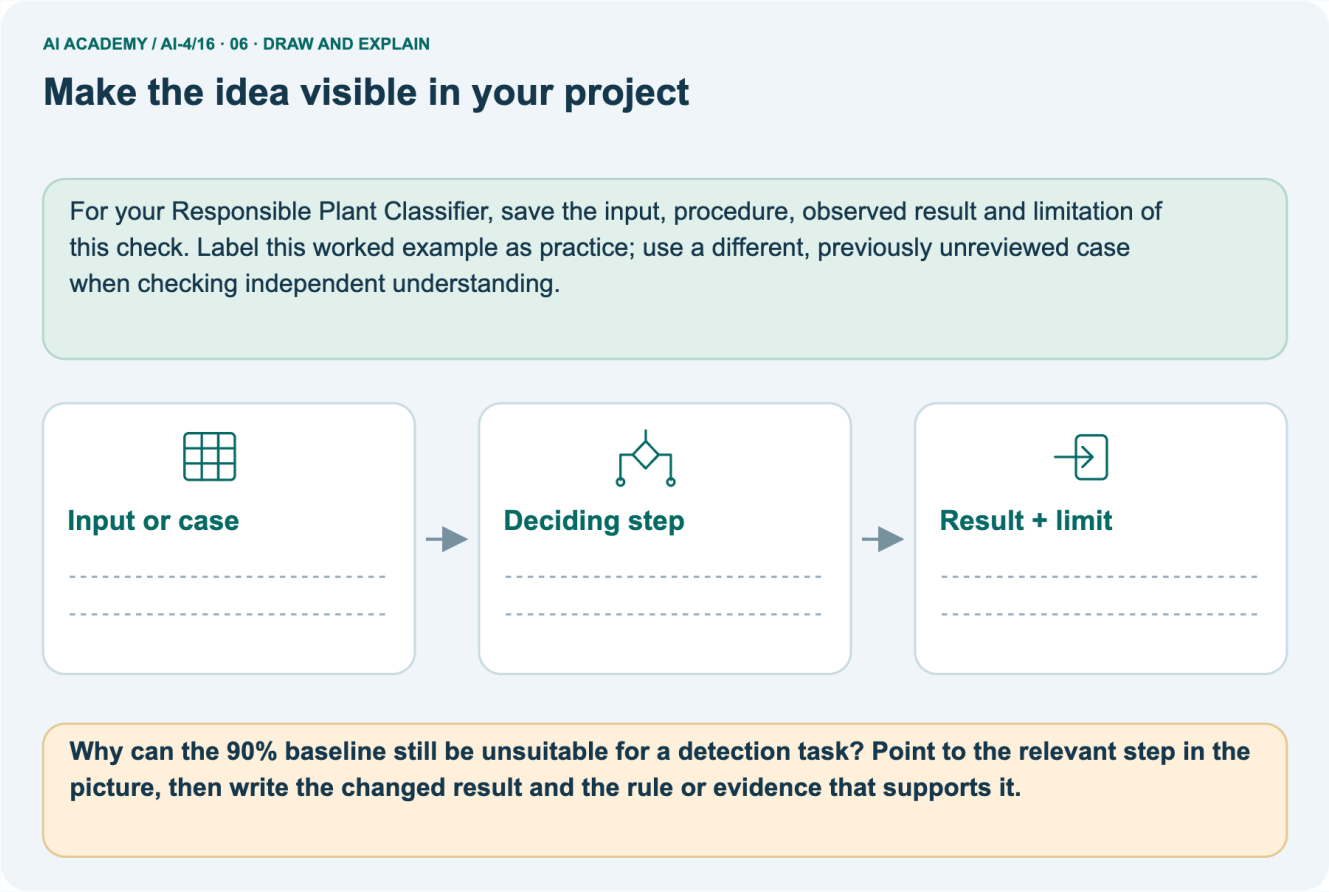
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Perfect performance on twenty supplied cases does not demonstrate dependable detection in other plants or settings.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why can the 90% baseline still be unsuitable for a detection task? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Accuracy and Imbalance: the case

02. Follow the deciding steps

03. Candidate decisions

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.